MOHAMED AHMED EL-TELEMY

Cairo, Egypt · +20 114 768 6582 · muhammed.eltelemy@gmail.com

PROFESSIONAL SUMMARY

BIM Automation Developer with a Civil Engineering background and specialized training in software development for the AEC industry through the Information Technology Institute (ITI) BIM Automation Developer Program. Experienced in developing Revit API add-ins, WPF desktop applications, and BIM automation solutions using C#, .NET, and MVVM architecture. Strong understanding of BIM workflows, object-oriented programming, database integration, software design principles, and automation technologies. Familiar with Generative AI concepts, prompt engineering, and AI-assisted workflows, with a strong interest in developing intelligent solutions that enhance productivity and decision-making within BIM environments. Passionate about building scalable software solutions that reduce manual work, improve data accessibility, and streamline engineering processes.

EDUCATION

Information Technology Institute (ITI) | BIM Automation Developer *2026 - Present*

- 460-hour intensive program focused on software development and BIM automation.
- Key topics: C#, .NET, OOP, Data Structures & Algorithms, Desktop Application Development, Revit API, AutoCAD API, Dynamo, IFC & xBIM, Database Fundamentals, Generative AI & Prompt Engineering.
- Approach: Project-based learning with real-world BIM automation solutions.

Mansoura University | Bachelor of Science in Civil Engineering *2018 – 2023*

- Graduation Project: Highway & Airport – Superpave Asphalt Mix Designs.
- Grade: Excellent.

Mansoura University | Preliminary MA. Public Works Engineering Department *2023 – present*

- Specified in Highway and Airport Engineering.
- Grade: Excellent.

Professional Experience

- **Jr. Highway Design Engineer** | PowerCem Technology Regional, Cairo *Jan 2024 – Jan 2026*
  specializing in advanced road reclamation (FDR) and diagnostic pavement evaluations

PROJECTS

RebarX | Revit API Add-in – Automated 3D Structural Reinforcement Detailing

- Developed an Autodesk Revit API add-in using C#, .NET Framework, and WPF/MVVM architecture to automate the generation of 3D reinforcement detailing for structural elements (isolated footings, columns, continuous/projected beams, flat slabs), reducing manual drafting time by up to 70%.
- Programmed parametric detailing rules compliant with the Egyptian Concrete Code (ECP 203), automating calculations for development lengths (Ld), tension/compression lap splices, hooks (90°, 135°, and 180°), and seismic confinement zones (Lo) in columns.
- Designed and implemented an intelligent commercial rebar slicing algorithm that automatically partitions longitudinal bars into standard 12-meter lengths, optimizing overlap splice locations, staggering splices, and minimizing scrap steel waste.
- Built an automated scheduling engine that generates structured, procurement-ready Bill of Materials (BOM) exports in CSV format, incorporating algorithms to sum quantities and prevent duplicated entries for repeated identical family instances.
- Created a custom Revit Ribbon panel with branded iconography, integrating asynchronous execution to keep Revit responsive, and implemented auto-minimization of the WPF user interface during generation to streamline user workflow.

Project Scorecard | Revit API Add-in – BIM Model Health Reports

- Developed a Revit API add-in using C# and .NET to generate automated BIM model health reports.
- Extracted model metadata including levels, categories, element counts, and project statistics.
- Improved model review efficiency by providing centralized project insights and performance indicators.
- Applied object-oriented programming principles to ensure maintainability and scalability.

Comment Stamper | Revit API Add-in – Bulk Parameter Management

- Designed and implemented a bulk parameter management tool using Autodesk Revit API.
- Automated comment and review status updates across large groups of model elements.
- Reduced repetitive manual operations and improved BIM coordination workflows.
- Enhanced productivity by streamlining model documentation processes.

Wall Data Export | Revit API Add-in – BIM Data Extraction

- Developed a BIM data extraction tool for collecting structured wall information from Revit models.
- Exported dimensions, types, levels, and related parameters to Excel for reporting and engineering analysis.
- Implemented reusable data processing components to support future automation solutions.
- Improved accessibility of project data for decision-making and quality control.

Building Inspection Tracker | WPF Desktop Application – MVVM Architecture

- Developed a desktop application using C#, WPF, and MVVM architecture.
- Implemented inspection tracking, checklist management, data filtering, and statistical reporting features.
- Applied software design principles and separation of concerns to improve maintainability.
- Designed a user-friendly interface to support engineering inspection workflows.

Company Management System | Desktop Application – EF Core & SQL Server

- Built a desktop management application using C#, Entity Framework Core, SQL Server, and SQLite.
- Developed full CRUD functionality with relational database integration.
- Implemented data validation and structured data management workflows.
- Applied layered architecture principles to support maintainability and future scalability.

TECHNICAL SKILLS

Programming & Development	BIM & AEC Development	Data & Databases	AI & Tools
C#, .NET, OOP, WPF	Revit API, AutoCAD API	SQL Server, SQLite	Generative AI, Prompt Engineering
MVVM, Desktop Development	Dynamo, IFC & xBIM	EF Core, CRUD Operations	Revit, AutoCAD, Visual Studio
Data Structures & Algorithms	BIM Automation Workflows	Database Design	Excel Integration, Windows

LANGUAGES

Arabic: Native | **English:** Professional Working Proficiency